

DATA PREPROCESSING AND FEATURE ENGINEERING REPORT

Customer Churn Prediction for ABC Communications Ltd

Machine Learning Internship Programme
AnalystLab Africa

Week 2 Assignment

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1. Project Overview

Customer churn prediction is an important machine learning application in the telecommunications industry. Customer churn occurs when an existing customer stops using a company's services.

For ABC Communications Ltd, predicting customers who are likely to churn can support proactive retention strategies, improve customer satisfaction, and reduce revenue losses.

Following the business understanding and exploratory data analysis completed during Week 1, the objective of Week 2 was to transform the raw Telco Customer Churn dataset into a clean and machine-learning-ready dataset.

The preprocessing workflow included data inspection, data cleaning, data type correction, feature engineering, categorical encoding, numerical scaling, outlier detection, feature selection, and preparation of training and testing datasets.

2. Business Objective

The overall business objective is to help ABC Communications Ltd identify customers who are at risk of leaving the company.

The machine learning system will eventually use customer demographic, service, contract, tenure, and billing information to predict whether a customer is likely to churn.

The Week 2 preprocessing stage ensures that these variables are appropriately prepared before machine learning models are developed.

The preprocessing process therefore aims to:

- Improve data quality.
- Correct inappropriate data types.
- Handle missing or invalid values.
- Remove irrelevant identifiers.
- Create meaningful additional features.
- Convert categorical variables into numerical representations.
- Scale numerical variables where appropriate.
- Investigate potential outliers.
- Identify useful features.
- Produce a consistent dataset suitable for model development.

3. Dataset Overview

The project uses the **Telco Customer Churn Dataset**.

The original dataset contains:

- 7,043 customer records
- 21 columns
- Demographic information
- Customer service information
- Contract information
- Payment information
- Billing information
- Churn status

The target variable is:

Churn

where:

- No = customer did not churn
- Yes = customer churned

The original dataset contains 20 predictor/identifier columns and one target variable

4. Initial Data Quality Assessment

The dataset was initially inspected using Python and Pandas.

The inspection included:

- Dataset dimensions
- Column names
- Data types
- Missing values
- Duplicate records
- Descriptive statistics
- Unique categorical values

The initial dataset shape was:

```
7,043 rows × 21 columns
```

The original dataset contained:

```
0 duplicate records
```

This indicated that there were no completely duplicated customer records requiring removal.

5. Data Type Correction

One important data quality issue was identified with the TotalCharges variable.

Initially:

TotalCharges = object

However, TotalCharges represents the total amount paid by a customer and should therefore be numerical.

The column was converted to a numeric data type.

After conversion:

TotalCharges → float64

This correction was necessary because numerical analysis, correlation analysis, scaling, and machine learning algorithms require the variable to be represented numerically.

6. Missing Value Treatment

Missing values were assessed using:

```
df.isnull().sum()
```

The raw dataset does not report conventional Pandas missing values because some blank values in TotalCharges are represented as whitespace.

Therefore, TotalCharges was converted using numeric coercion so that invalid blank entries could be identified.

These values were subsequently treated appropriately during preprocessing.

After preprocessing, the final processed dataset contained:

```
Missing values: 0
```

This confirms that the machine-learning-ready dataset contains no remaining missing values.

7. Duplicate Record Assessment

Duplicate records were checked using `Pandas duplicated()` function.

The original dataset contained:

0 duplicate records

Therefore, no records were removed at the raw dataset stage.

After removing customerID and transforming the features, **22 identical processed feature representations** were observed.

These were not automatically removed because they do not necessarily represent duplicate customers. Different customers can legitimately have identical demographic, service, tenure, and billing characteristics.

Therefore, the 22 processed duplicate representations were retained.

This avoids accidentally removing valid customer observations.

8. Removal of Irrelevant Features

The customerID variable was removed from the dataset.

Reason

customerID is a unique identifier rather than a meaningful predictor of customer churn.

Including it could cause the model to learn patterns associated with arbitrary identifiers rather than actual customer behavior.

After removing customerID, the dataset changed from:

21 columns to 20 columns

This was an intentional feature-removal decision.

9. Feature Engineering

Feature engineering was performed to create additional variables that may provide useful information for predicting customer churn.

Two new features were created.

9.1 Tenure Group

A new variable called:

`tenure_group` was created from the original tenure variable.

The feature groups customers according to their length of relationship with the company.

The groups used were:

- New
- Developing
- Established
- Long-term

Reason

Customer tenure is potentially related to customer loyalty and churn risk.

Grouping tenure provides an alternative representation that can help identify differences between newer and longer-term customers.

10. Total Services

A second engineered variable called: `total_services` was created.

This represents the number of relevant services used by each customer.

For example, a customer using several additional services receives a higher `total_services` value.

Reason

The number of services a customer uses may provide an indication of customer engagement with the company.

A customer who uses multiple services may behave differently from a customer using only one service.

Therefore, this feature was included as a potentially useful predictor of churn.

11. Categorical Feature Encoding

The dataset contains numerous categorical variables, including:

- Gender
- Partner
- Dependents
- PhoneService
- MultipleLines
- InternetService
- Contract
- PaymentMethod
- PaperlessBilling
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies

Machine learning algorithms generally require numerical input.

Therefore, categorical variables were transformed using **One-Hot Encoding**.

One-Hot Encoding creates separate binary columns representing the possible categories.

For example: [Contract contains: Month-to-month, One year, Two year](#)

These categories can be represented as separate numerical variables.

Why One-Hot Encoding?

One-Hot Encoding was selected because most of the categorical variables are nominal rather than naturally ordered.

For example:

- Electronic check
- Mailed check
- Credit card
- Bank transfer

do not have a meaningful numerical order.

Assigning values such as:

- Electronic check = 1
- Mailed check = 2
- Credit card = 3

could incorrectly imply an ordinal relationship.

One-Hot Encoding avoids this problem.

12. Numerical Feature Scaling

The main numerical variables include:

- SeniorCitizen
- tenure
- MonthlyCharges
- TotalCharges
- total_services

Numerical features were scaled using:

[StandardScaler](#)

Standardization transforms numerical variables so that their distributions are represented on a comparable scale.

The transformation is based on:

$$z = \frac{x - \mu}{\sigma}$$

where:

- x = original value
- μ = mean
- σ = standard deviation

Why StandardScaler?

StandardScaler was selected because the numerical variables have substantially different ranges.

For example:

tenure = 0–72

MonthlyCharges = approximately 18–119

TotalCharges = substantially larger range

Without scaling, variables with larger numerical magnitudes could disproportionately influence algorithms that are sensitive to feature scales.

13. Outlier Detection

Potential outliers were investigated using the **Interquartile Range (IQR)** method.

The IQR is calculated as: The lower and upper bounds are:

The following numerical features were investigated:

- tenure
- MonthlyCharges
- TotalCharges
- total_services

The analysis identified

Tenure: 0 outliers

$$IQR = Q3 - Q1$$

$$Lower = Q1 - 1.5(IQR)$$

$$Upper = Q3 + 1.5(IQR)$$

MonthlyCharges: 0 outliers

TotalCharges: 0 outliers

total_services: 0 outliers

Therefore, no observations were removed using the IQR method.

This was an important decision because removing legitimate high-value customers or customers with high charges could potentially remove useful information from the dataset.

14. Feature Selection

Feature selection was performed using several analytical approaches.

These included:

- Correlation analysis
- Variance analysis
- Domain reasoning

The correlation analysis provided information about relationships between numerical features and the target.

For example:

tenure → Churn: -0.352

MonthlyCharges → Churn: 0.193

TotalCharges → Churn: -0.198

SeniorCitizen → Churn: 0.151

total_services → Churn: approximately -0.019

Interpretation

tenure showed a negative relationship with churn, suggesting that customers with longer tenure tend to have lower churn rates.

MonthlyCharges showed a positive relationship with churn, indicating that higher monthly charges may be associated with increased churn risk.

However, correlation alone does not establish causation and does not fully determine feature importance.

Therefore, these findings will be supplemented with model-based feature importance during future model development.

15. Feature Variance

Variance was also examined to identify features with very little variation.

The numerical feature variances included approximately:

Feature	Variance
SeniorCitizen	0.136
tenure	603.168
MonthlyCharges	905.411
TotalCharges	5,138,357
total_services	5.349

The analysis did not identify a feature with zero variance that would provide no information.

Therefore, no major numerical feature was removed solely because of zero variance.

16. Train-Test Split

The dataset was divided into training and testing subsets.

An **80/20 split** was used.

The resulting datasets were:

Dataset	Rows	Processed Features
Training	5,634	50
Testing	1,409	50

The split was stratified according to the target variable to preserve the approximate churn distribution in both datasets.

This is important because the target variable is imbalanced:

No Churn: 73.46%

Churn: 26.54%

Stratification helps ensure that the training and testing sets maintain similar class proportions.

17. Final Processed Dataset

After preprocessing, the complete dataset contains:

Rows: 7,043

Columns: 51

The 51 columns consist of:

- 50 processed predictor features
- 1 target variable
- The final dataset contains:

Missing values: 0

The target variable was encoded as:

- 0 = No Churn
- 1 = Churn

The final processed dataset was saved as:

processed_telco_churn.csv

18. Preprocessing Summary

Preprocessing Step	Action Taken
Missing values	Identified and treated
Duplicate records	Original duplicates checked; none found
customerID	Removed
TotalCharges	Converted to numeric
tenure_group	Created
total_services	Created
Categorical variables	One-Hot Encoded
Numerical variables	StandardScaler applied
Outliers	IQR analysis performed
Feature selection	Correlation, variance and domain reasoning
Target	Encoded as 0/1
Train/test split	80/20 stratified
Final missing values	0
Final processed features	50

19. Machine-Learning Readiness

Following preprocessing, the dataset is considered ready for machine learning model development.

The processed data:

- Contains numerical features.
- Contains no remaining missing values.
- Has categorical variables encoded.
- Has numerical variables scaled.

- Has irrelevant customer identifiers removed.
- Contains engineered features.
- Has been separated into training and testing sets.
- Has a clearly defined binary target.

20. Conclusion

The Week 2 preprocessing stage successfully transformed the raw Telco Customer Churn dataset into a machine-learning-ready representation.

The process addressed key data quality issues, including the incorrect data type of TotalCharges, missing or invalid numerical entries, irrelevant identifiers, categorical variables, feature-scale differences, and potential outliers.

Feature engineering produced tenure_group and total_services, providing additional representations of customer tenure and service engagement.

Categorical variables were transformed using One-Hot Encoding, while numerical variables were standardized using StandardScaler.

Outlier analysis using the IQR method did not identify significant outliers requiring removal.

Correlation and variance analysis were used to support feature-selection decisions.

The final processed dataset contains **7,043 customer records, 50 processed predictor features, and one binary target variable**, with **zero remaining missing values**.

The processed dataset is therefore ready for the next stage of the project: **machine learning model development and evaluation**.